

HIRAM ZUNIGA

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SUMMARY

Software engineer (B.S. Computer Science, 2026) focused on backend and full-stack development, cloud services, REST APIs, databases, and deployed web applications. Hands-on experience building event-driven AWS systems, PostgreSQL backend services, API integrations, dashboards (including Power BI), and production automation pipelines. Bilingual Spanish/English, with a TOEFL iBT score of 100/120.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java Fundamentals

Backend & APIs: FastAPI, REST API design and integration, PostgreSQL, AWS RDS, DynamoDB, API Gateway

Cloud & Systems: AWS Lambda, Step Functions, SQS, SNS, S3, RDS, DynamoDB, serverless architecture, Linux

Applications & Tools: Streamlit, HTML/CSS, Power Apps, Power BI, Git/GitHub, unit testing, data validation

EXPERIENCE

Software Engineering Mentee – Message Router

Aug 2025 – January 2026

AWS Mentorship Program

Tijuana, MX

- Designed and built a serverless message-routing platform using API Gateway, EventBridge, SNS, SQS, Lambda, Step Functions, PostgreSQL/RDS, DynamoDB, and S3.
- Implemented backend reliability patterns including retries, dead-letter queues, and scheduled event flows.
- Demoed the system to AWS engineers at the **Amazon San Diego office** as part of a software engineering mentorship program.

Data Analyst – Software Automation Focus

Mar 2025 – Sep 2025

I3PL

Tijuana, MX

- Automated daily reporting by integrating multiple REST APIs into scheduled Python pipelines publishing to SharePoint, **cutting manual work by 80%**.
- Built validation rules, monitoring checks, and standardized data flows that reduced errors and rework across company-wide logistics operations.
- Delivered SQL and Power BI **dashboards** for real-time milestone tracking, reducing container dwell time (>2 hours per load) through proactive alerts.

Research Assistant – Software Systems

Oct 2025 – Present

CETYS University – Advisor: Dr. Ulises Orozco-Rosas

Tijuana, MX

- Built end-to-end **software pipelines** for data ingestion, preprocessing, evaluation, inference, documentation, and demos across computer vision and robotics projects.
- Integrated backend logic, data processing modules, and user-facing application workflows into deployable prototypes for applied AI and robotics systems.
- Created reproducible validation workflows for system benchmarking, documentation, and demos.

SELECTED SOFTWARE PROJECTS

StudyBot – LLM-Powered Study Assistant

2025

Personal Project

Project Demo

- Built a user-facing study assistant with retrieval-augmented generation (RAG) workflows for document summarization, quiz generation, and question answering over user-provided material.

OpenHealth – Deployed Resource Platform

Nov 2025

SDSU Big Data Hackathon

Project Page

- Built and deployed a web platform connecting users to nearby shelters, weather alerts, QR-based health summaries, and mental-health support resources.

AWARDS & ACCEPTED PRESENTATIONS

3rd place, MIT iQuHACK 2026 · **3rd place**, Yale YQuantum 2026 · **Regional 1st place**, NASA Space Apps 2025 · **2nd place**, GEAR Web3 Hackathon 2023 (**\$1,000 USD prize**) · **NVIDIA** Certificate: Getting Started with Deep Learning.
Accepted oral presentations: SPIE Optics + Photonics 2026 (×2), San Diego, CA; KES 2026, **Ireland, Europe**.

EDUCATION

CETYS University

2022 – 2026

B.S. in Computer Science Engineering

Tijuana, MX

- Relevant coursework: Data Structures, Algorithms, Object-Oriented Programming, Databases, Parallel Computing, Machine Learning, Deep Learning, Computer Vision, Probability & Statistics.

Yonsei University – Winter Abroad Program

Dec 2023 – Jan 2024

Big Data Analysis

Seoul, South Korea